

Discrete optimization

Set covering

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Optimization: principles and algorithms



Set covering

We review some classical combinatorial optimization problems and show how they can be modeled as a (mixed) integer linear optimization problem.

In this video, we introduce the set covering problem

Set covering



The set covering problem

- ▶ After the FIFA World Cup, Camille wants to complete her collection of stickers.
- ▶ She can buy collections of stickers from her schoolmates.
- ▶ In each collection, there are stickers that she needs, but also stickers that she does not need.
- ▶ The schoolmates do not accept to sell stickers individually. The whole collection has to be purchased.
- ▶ Camille must decide which collections to purchase, in order to complete her own album, at a minimum price.



Definition

Data

- ▶ A set U of m elements.
- ▶ $S_i \subseteq U$, $i = 1, \dots, n$.
- ▶ $a_{ij} = 1$ if element j belongs to subset S_i .
- ▶ Costs: c_i .

Objective

Choose J subsets S_{i_j} , $j = 1, \dots, J$, of minimal total cost such that

$$\bigcup_{j=1}^J S_{i_j} = U.$$

Modeling

Decision variables

$$x_i = \begin{cases} 1 & \text{if subset } i \text{ is selected ,} \\ 0 & \text{otherwise} \end{cases}$$

Objective function

$$\min f(x) = \sum_{i=1}^n c_i x_i$$

Constraints

$$\sum_{i=1}^n a_{ij} x_i \geq 1, j = 1, \dots, m \quad x_i \in \{0, 1\}, i = 1, \dots, n$$

Summary

The set covering problem is also a classical combinatorial optimization problem, relevant in many contexts, such as personnel shift planning, for instance. We have shown that it can be written as an integer optimization problem.